

Miad Ahmadi

Electrical Engineer

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Summary

Electrical Engineer with a Ph.D. in Power Electronics and Reliability, currently working at Hitachi Energy on grid automation systems. Skilled in HIL testing, control and protection systems, and PCB design. Strong academic and industry background in fault-tolerant converter topologies, reliability modeling, and energy systems.

Experience

Control & Protection Engineer, Hitachi Energy, The Netherlands Feb 2025 – Present

- Developing grid automation and protection solutions using IEDs (REF650) in PCM600.
- Converting and verifying protection logic for high-voltage substations.
- Integrating control systems for energy infrastructure and AC substations.

Ph.D. Researcher in Power Electronics and Reliability, TU Delft, The Netherlands Sept 2021 – Jan 2025

- Developed reliability-centered maintenance, failure rate analysis, and CAPEX/OPEX-aware designs.
- Supervised M.Sc. students; contributed to industrial collaboration and teaching.
- Designed power electronic systems and PCBs (Schematics & Layout in Altium).
- Contributed to fault-tolerant MMC converters using HIL and hardware testing.
- Carried out reliability and thermal management analysis per industry standards.
- Led experimental validation using high-voltage benches and measurement tools.

Research Fellow, Polytechnic University of Milan, Italy Apr 2020 – Sept 2021

- Designed power flow control for railway electrification using renewables.
- Built a Digital Twin for 9kV DC railway grid behavior prediction.
- Modeled DC-DC and AC-DC converters; developed variable frequency inverters.

Power Systems Reliability Intern, SuperGrid Institute, France Apr 2019 – Oct 2019

- Defined functional and protection requirements for MTDC systems.
- Simulated DC faults and DCCB capability; used EMTP simulations for layout optimization.

Education

Ph.D. in Electrical Engineering, Delft University of Technology, The Netherlands 2021 – 2025

Thesis: *Reliability of Modular Multilevel Converters: Impact of Redundancy, Modularity and Reconfigurability*

M.Sc. in Electrical Engineering, Polytechnic University of Milan, Italy 2017 – 2019

Grade: 110/110 | Thesis: *Reliability Analysis of the Bus-Bar Systems in MMC VSC MTDC*

B.Sc. in Electronics Engineering, Razi University, Iran 2012 – 2016

Grade: 16.03/20 | Thesis: *Effect of Impedance Matching on Wireless Power Transfer Systems*

Skills

Technical: MATLAB/SIMULINK, Python, OPAL-RT, Altium Designer, FPGA, Microcontrollers, HIL Testing, MIL-HDBK 217, Weibull, FMECA, Monte Carlo Simulation, PCM600, IEC 61850, IED configuration (REF630/650).
Transferable: Teamwork, Leadership, Analytical Thinking, Problem Solving, Time Management, Communication, Creativity, Willingness to Learn.

Honors & Awards

- **IEEE IES SYPA Grant** – 1st winner, CPE POWERENG Conference, 2023
- **IEEE Best Paper Award** – Paper on modularity and redundancy, 2022
- **Project Fellowship** – Polytechnic University of Milan, 2020
- **Top 5% GPA** – Among Electrical Engineering graduates, Polimi, 2019